CPSC 4900 Senior Project

ProtSCAPE: Mapping the landscape of protein conformations in molecular dynamics

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Introduction

ProtSCAPE is a deep learning framework designed to map and generate the full range of protein conformations observed in **molecular dynamics (MD) simulations,** to capture realistic structural flexibility.

We evaluate whether ProtSCAPE can reproduce MD ensembles in terms of metrics such as **RMSD, Wasserstein distance, and RMSF correlation.**

Results demonstrate physically plausible protein structures while offering a promising approach for **upsampling** of complex protein motion landscapes.

ProtSCAPE KEY FINDINGS

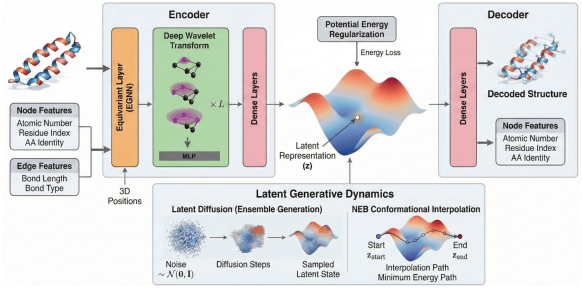
> ProtSCAPE achieved the **best overall ensemble agreement** on 4 of 5 proteins in distributional metrics (RMWD and PCA-W_2)

> **RMSF correlations were ~0.999** across all systems, showing near-perfect recovery of residue flexibility patterns.

> Results support ProtSCAPE as an effective method for **upsampling sparse MD trajectories** into accurate full ensembles.

ProtSCAPE Architecture

ProtSCAPE uses an encoder-decoder architecture in which protein structures are represented as **graphs** and processed through an equivariant geometric network.. It combines graph wavelet transforms with a **latent diffusion model** to learn a smooth, **energy-aware latent space.**



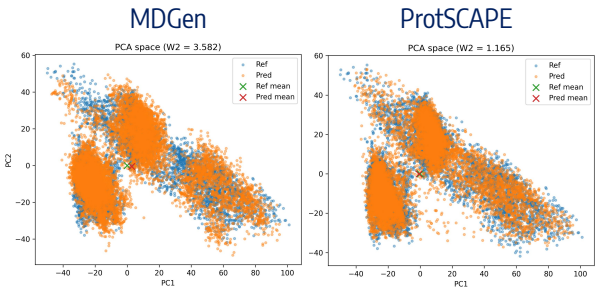
Methodology and Metrics

Upsampling refers to training on a **sparse subset** of MD frames and generating a larger ensemble that **reconstructs the full conformational landscape** at lower computational cost.

We use the **ATLAS** dataset, with multiple **100 ns** simulation replicates across representative protein systems.

Model performance is evaluated using structural validity metrics such as **MolProbity scores** and ensemble agreement metrics including **RMSD, RMSF correlation, and Wasserstein-based distances.**

ProtSCAPE was trained on **subsampled ATLAS trajectories** and benchmarked against other models, including **AlphaFLOW and MDGen**, under the same upsampling conditions.



PCA Plots of C-alpha atoms for predicted ensemble (orange) and reference MD simulations (blue) for system 6H86

Results

In ensemble-level comparisons, ProtSCAPE closely reproduced residue flexibility patterns observed in MD, with **RMSF correlations near 1.0.** Pairwise RMSD and PCA analyses further showed that ProtSCAPE captured global spread and collective conformational motions

Although MDGen achieved lower Wasserstein distances on several systems, ProtSCAPE remained competitive while producing **smoother sampling and stronger structural correctness.** Together, these results support ProtSCAPE as an effective framework for efficient upsampling of MD ensembles.

Protein	Method	RMWD↓	Trans.↓	Var.↓	Pool PCA-W_2 ↓	Ref PCA-W_2 ↓
6H86 (Chain A)	AlphaFlow	4.2487 ± 0.017	2.4578 ± 0.005	3.4657 ± 0.021	34.9194 ± 0.195	34.6883 ± 0.195
	MDGen	1.0066 ± 0.008	0.8097 ± 0.008	0.5978 ± 0.014	3.8805 ± 0.209	3.7751 ± 0.207
	ProtSCAPE	$\mathbf{0.3441 \pm 0.016}$	$\mathbf{0.2070 \pm 0.013}$	$\mathbf{0.2749 \pm 0.010}$	$\mathbf{1.4490 \pm 0.215}$	$\mathbf{1.4576 \pm 0.215}$
6P5H (Chain A)	AlphaFlow	1.0504 ± 0.061	0.7488 ± 0.077	0.7350 ± 0.010	8.0587 ± 0.839	8.0806 ± 0.830
	MDGen	0.6200 ± 0.009	0.4258 ± 0.013	0.4504 ± 0.015	2.2015 ± 0.131	2.0740 ± 0.140
	ProtSCAPE	$\mathbf{0.2127 \pm 0.005}$	$\mathbf{0.1424 \pm 0.006}$	$\mathbf{0.1579 \pm 0.002}$	$\mathbf{0.9370 \pm 0.125}$	$\mathbf{0.9449 \pm 0.125}$
6JV8 (Chain A)	AlphaFlow	1.1663 ± 0.012	0.6576 ± 0.015	0.9631 ± 0.007	7.8291 ± 0.179	7.8324 ± 0.175
	MDGen	0.4729 ± 0.014	0.3818 ± 0.017	0.2787 ± 0.011	2.4517 ± 0.182	2.4535 ± 0.188
	ProtSCAPE	$\mathbf{0.1980 \pm 0.003}$	$\mathbf{0.1298 \pm 0.004}$	$\mathbf{0.1494 \pm 0.003}$	$\mathbf{0.4991 \pm 0.067}$	$\mathbf{0.5173 \pm 0.064}$
7LP1 (Chain A)	AlphaFlow	1.2303 ± 0.017	0.7533 ± 0.013	0.9727 ± 0.013	3.8058 ± 0.153	3.7298 ± 0.140
	MDGen	0.4694 ± 0.014	0.3623 ± 0.014	0.2983 ± 0.009	0.8995 ± 0.180	1.1783 ± 0.111
	ProtSCAPE	$\mathbf{0.1916 \pm 0.010}$	$\mathbf{0.0847 \pm 0.009}$	$\mathbf{0.1718 \pm 0.007}$	$\mathbf{0.3511 \pm 0.091}$	$\mathbf{0.3594 \pm 0.091}$
7JFL (Chain C)	AlphaFlow	1.9544 ± 0.050	1.5941 ± 0.028	1.1305 ± 0.048	13.0873 ± 0.352	13.0587 ± 0.351
	MDGen	0.3431 ± 0.003	0.2088 ± 0.007	0.2721 ± 0.003	$\mathbf{0.5640 \pm 0.043}$	$\mathbf{0.5820 \pm 0.041}$
	ProtSCAPE	$\mathbf{0.2117 \pm 0.006}$	$\mathbf{0.1691 \pm 0.006}$	$\mathbf{0.1273 \pm 0.003}$	0.7591 ± 0.070	0.7569 ± 0.071

Above: Distributional ensemble quality metrics comparing generated protein conformations. Lower values indicate better agreement for RMWD, its translation and variance contributions, and W_2 distances in pooled or reference PCA spaces.

Right: Dynamic agreement metrics evaluating how well generated ensembles reproduce flexibility. Higher RMSF correlation indicates more accurate recovery of residue-level fluctuation patterns; lower pairwise RMSD indicates closer agreement in ensemble spread and conformational variability.

Protein	Method	RMSF Corr.↑	Pairwise RMSD↓
6H86 (Chain A)	AlphaFlow	0.8965 ± 0.013	0.7875 ± 0.004
	MDGen	0.9964 ± 0.000	0.1141 ± 0.018
	ProtSCAPE	$\mathbf{0.9997 \pm 0.000}$	$\mathbf{0.0664 \pm 0.010}$
6P5H (Chain A)	AlphaFlow	0.9980 ± 0.000	0.2234 ± 0.006
	MDGen	0.9979 ± 0.000	0.1082 ± 0.016
	ProtSCAPE	$\mathbf{0.9998 \pm 0.000}$	$\mathbf{0.0403 \pm 0.015}$
6JV8 (Chain A)	AlphaFlow	0.9971 ± 0.000	0.3845 ± 0.007
	MDGen	0.9986 ± 0.000	$\mathbf{0.0332 \pm 0.011}$
	ProtSCAPE	$\mathbf{0.9994 \pm 0.000}$	0.0458 ± 0.012
7LP1 (Chain A)	AlphaFlow	0.9548 ± 0.003	0.4331 ± 0.012
	MDGen	0.9932 ± 0.000	$\mathbf{0.0293 \pm 0.010}$
	ProtSCAPE	$\mathbf{0.9988 \pm 0.000}$	0.0701 ± 0.008
7JFL (Chain C)	AlphaFlow	0.9888 ± 0.002	0.4351 ± 0.013
	MDGen	0.9985 ± 0.000	0.0448 ± 0.008
	ProtSCAPE	$\mathbf{0.9998 \pm 0.000}$	$\mathbf{0.0426 \pm 0.009}$

Conclusions & Future Work

ProtSCAPE demonstrates that learned latent diffusion models can efficiently **reconstruct realistic protein conformational** ensembles from **sparsely sampled** MD data. By combining strong structural validity with accurate recovery of dynamic behavior, the model offers a promising tool for reducing the computational cost of protein simulation analysis.

Future work will focus on improving distributional agreement with MD ensembles, expanding evaluation to **larger and more flexible** protein systems, and applying ProtSCAPE to biologically relevant **transition pathways** such as ligand binding and folding events.